

Lophotrochozoans, a clade identified by molecular data, have the widest range of animal body forms



The vast majority of animal species belong to the clade Bilateria, whose members exhibit bilateral symmetry and are triploblastic (see

Concept 32.3). Most bilaterians also have a digestive tract with two openings (a mouth and an anus) and a body cavity (a coelom or hemocoel; see Figure 32.9). Recent DNA analyses suggest that the common ancestor of living bilaterians lived about 670 million years ago. To date, however, the oldest fossil that is widely accepted as a bilaterian is of *Kimberella*, a mollusc (or close relative) that lived 560 million years ago (see Figure 32.5b). Many other bilaterian groups first appeared in the fossil record during the Cambrian explosion (535 to 525 million years ago).

Molecular evidence suggests that today there are three major clades of bilaterally symmetrical animals: Lophotrochozoa, Ecdysozoa, and Deuterostomia. This section will focus on the first of these clades, the lophotrochozoans. Concepts 33.4 and 33.5 will explore the other two clades.

Although the clade Lophotrochozoa was identified by molecular data, its name comes from features found in some of its members. Some lophotrochozoans develop a structure called a *lophophore*, a crown of ciliated tentacles that functions in feeding, while others go through a distinctive stage called the *trochophore larva* (see Figure 32.12). Other members of the group have neither of these features. Few other unique morphological features are widely shared within the group—in fact, the lophotrochozoans are the most diverse bilaterian clade in terms of body plan. This diversity in form is reflected in the number of phyla classified in the group: Lophotrochozoa includes 18 phyla, more than twice the number in any other clade of bilaterians.

We'll now introduce six of the diverse lophotrochozoan phyla: the flatworms, rotifers and acanthocephalans, ectoprocts, brachiopods, molluscs, and annelids.

Flatworms

Flatworms (phylum Platyhelminthes) live in marine, freshwater, and damp terrestrial habitats. In addition to free-living species, flatworms include many parasitic species, such as flukes and tapeworms. Flatworms are so named because they have thin bodies that are flattened dorsoventrally (between the dorsal and ventral surfaces); the word *platyhelminth*

means “flat worm.” (Note that *worm* is not a formal scientific name but rather refers to animals with long, thin bodies.) The smallest flatworms are nearly microscopic free-living species, while some tapeworms are more than 20 m long.

Although flatworms are triploblastic, they lack a body cavity. Their flat shape increases their surface area, placing all their cells close to water in the surrounding environment or in their gut. Because of this proximity to water, gas exchange and the elimination of nitrogenous waste (ammonia) can occur by diffusion across the body surface. As shown in **Figure 33.8**, a flat shape is one of several structural features that maximize surface area and have arisen (by convergent evolution) in different groups of animals and other organisms.

As you might expect since all their cells are close to water, flatworms have no organs specialized for gas exchange, and their relatively simple excretory apparatus functions mainly to maintain osmotic balance with their surroundings. This apparatus consists of **protonephridia**, networks of tubules with ciliated structures called *flame bulbs* that pull fluid through branched ducts opening to the outside (see Figure 44.8). Most flatworms have a gastrovascular cavity with only one opening. Though flatworms lack a circulatory system, the fine branches of the gastrovascular cavity distribute food directly to the animal's cells.

Early in their evolutionary history, flatworms separated into two lineages, Catenulida and Rhabditophora. Catenulida is a small clade of about 100 flatworm species, most of which live in freshwater habitats. Catenulids typically reproduce asexually by budding at their posterior end. The offspring often produce their own buds before detaching from the parent, thereby forming a chain of two to four genetically identical individuals—hence their informal name, “chain worms.”

The other ancient flatworm lineage, Rhabditophora, is a diverse clade of about 20,000 freshwater and marine species, one example of which is shown in Figure 33.8. We'll explore the rhabditophorans in more detail, focusing on free-living and parasitic members of this clade.

Free-Living Species

Free-living rhabditophorans are important as predators and scavengers in a wide range of freshwater and marine habitats. The best-known members of this group are freshwater species in the genus *Dugesia*, commonly called **planarians**. Abundant in unpolluted ponds and streams, planarians prey on smaller animals or feed on dead animals. They move by using cilia on their ventral surface, gliding along a film of mucus they secrete. Some other rhabditophorans also use their muscles to swim through water with an undulating motion.

A planarian's head features a pair of light-sensitive eyespots as well as lateral flaps that function mainly to detect specific chemicals. The planarian nervous system is more complex and centralized than the nerve nets of cnidarians